



Las Vegas has had unusually dry monsoon season



A monsoon storm cell emits lightning over the southern part of the Las Vegas valley Friday, July 20, 2018. Photo by: Sam Morris/Las Vegas News Bureau

By [Ilana Williams](#) ([contact](#))

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The monsoon season and the rain it usually produces has been abnormally dry this summer in Las Vegas.

The season, which runs from June through mid-September, has dropped just 0.08 inches of rain here, according to the National Weather Service.

By comparison, the monsoon seasons from 1991 to 2020 dumped an average of 1.06 inches of rain annually, the weather service said.

A monsoon is a seasonal reversal of winds where one season has winds blowing from the ocean onto land, which generally results in a lot of rainfall, and another season in which winds blowing from land over the ocean causes a dry season, said Matthew Lachniet, professor of geology at UNLV.

Although the summer hasn't seen as much rain as normal, it was still monsoon season, said Mark Austin, a meteorologist at the National Weather Service in Las Vegas.

"It's just been pretty abysmal in terms of rainfall (for) monsoon season," Woods said. "It's been below normal precipitation for our whole area."

Summer rainfall isn't always a product of a monsoon. Take last summer, when Las Vegas received 1.17 inches of rain in August and 1.15 inches in September because [of Tropical Storm Hilary](#).

Yet, the storm was not considered part of monsoon season because the winds affected the area differently, Lachniet said.

"If you were to look at the summer rainfall totals, they would have been quite high during Hurricane Hilary because it rained a lot," Lachniet said. "It's a hurricane that turned into a tropical storm that happened to get steered into the Southwest by some unusual wind patterns."

In the Southwestern U.S., the summer monsoon carries moist air from the Gulf of California north along the Colorado River Valley. Traditionally, Arizona and New Mexico experience more monsoon rain in June while Nevada gets rain in July.

Warm sea surface temperatures cause water to evaporate, Lachniet said. As humid air lifts higher in the atmosphere, it cools off and causes rain.

Southern Nevada gets a strong monsoon season when the winds blow from south to north, Lachniet said. And in some years, most recently 2020, there was no measurable rainfall during the season.

What drives the monsoon to occur is a zone of high-pressure air that spins clockwise, over the four corners of the Southwest, Lachniet said. By being southwest of the high-pressure zone, winds from the Gulf of California pull in more moisture.

However, if the high-pressure zone moves west over Nevada, it does not allow moisture from the Gulf of California.

The summer [brought record-setting temperatures](#), hitting 120 degrees in Las Vegas on July 7 to surpass the previous all-time high of 117 degrees. Las Vegas also had more than 100 consecutive days of temperatures over 100 degrees.

“Climate change has definitely increased temperatures already in the Southwest,” Lachniet said.
“Warmer ground temperature causes more of that rainfall to just evaporate back up into the atmosphere.”

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